Self-Singed Certificate HTTPS server

1) HTTP Server

Step 1. Apache 패키지 설치

❶ 패키지 설치 및 확인

yum install –y httpd

rpm –qa | grep httpd

```
[root@open /]# rpm -qa | grep httpd
httpd-tools-2.4.6-40.el7.centos.4.x86_64
httpd-2.4.6-40.el7.centos.4.x86_64
```

❷ 패키지 버전 확인

#httpd –v

```
[root@open httpd]# httpd -v
Server version: Apache/2.4.6 (CentOS)
Server built:   Jul 18 2016 15:30:14
```

Step 2. OpenSSL와 mod_ssl 설치

❶ openssl 패키지 설치

yum –y install openssl-*

rpm –qa |grep openssl

```
[root@open modules]# rpm -qa | grep openssl
openssl-libs-1.0.1e-51.el7_2.7.x86_64
openssl-static-1.0.1e-51.el7_2.7.x86_64
openssl098e-0.9.8e-29.el7.centos.2.x86_64
openssl-perl-1.0.1e-51.el7_2.7.x86_64
openssl-devel-1.0.1e-51.el7_2.7.x86_64
openssl-1.0.1e-51.el7_2.7.x86_64
```

❷ mod_ssl 모듈 확인

```
[root@open /]# httpd -l
Compiled in modules:
  core.c
  mod_so.c
  http_core.c
```

[동적으로 설치된 mod_ssl module]

❸ 동적인 방식의 경우 mod_so.c 를 확인 후 module 디렉터리 내에 mod_ssl.so 이 있는지 확인

#ls /etc/httdp/modules/mod_ssl.so

mod_ssl.so 파일이 없는 경우

rpm –qa |grep mod_ssl

#yum –y install mod_ssl*

#ls /etc/httdp/modules/mod_ssl.so

```
[root@localhost /]# ls /etc/httpd/modules/mod_ssl.so
/etc/httpd/modules/mod_ssl.so
```

Step 3. httpd.conf 파일 확인 후 index.html 파일 만들기

vi /etc/httpd/conf/httpd.conf

```
119 DocumentRoot "/var/www/html"
```

vi /var/www/html/index.html

```
<html>
<head>
<title> OPNE.NET Page </title>
</head>
<body>
This is Our HoMEPage^^
</body>
</html>
```

Step 4. http 프로토콜 방화벽 설정

#iptables –I INPUT –m state --state NEW, ESTABLISHED –p tcp --dport 80 -j ACCEPT

#service iptables save

Step 5. http 서비스 활성화

#service iptables start

#service httpd start

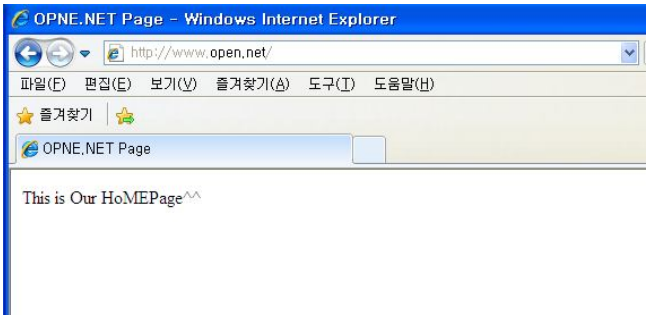
Step 6. 클라이언트 확인

❶ C:\>windows>system32>drivers>etc>

- Host파일 메모장이나 워드패드로 열기 도메인 명 지정

192.168.10.30 www.open.net

❷ http://www.open.net



2) HTTPS Server 구성

Step 1. 개인키 생성

#cd /etc/pki/tls/certs

#openssl genrsa –out www.open.net.key 2048

```
[root@open certs]# openssl genrsa -out open.net.key 2048
Generating RSA private key, 2048 bit long modulus
.........................+++
...............+++
e is 65537 (0x10001)
```

Step 2. CSR 생성

#openssl req –new –key www.open.net.key -out www.open.net.csr

*Common Name : www.open.net

```
[root@open certs]# openssl req -new -key open.net.key -out open.net.csr
You are about to be asked to enter information that will be incorporated
into your certificate request.
What you are about to enter is what is called a Distinguished Name or a DN.
There are quite a few fields but you can leave some blank
For some fields there will be a default value,
If you enter '.', the field will be left blank.
-----
Country Name (2 letter code) [XX]:KR
State or Province Name (full name) []:Seoul
Locality Name (eg, city) [Default City]:Seoul
Organization Name (eg, company) [Default Company Ltd]:OpenSouce Corp.
Organizational Unit Name (eg, section) []:Security&System
Common Name (eg, your name or your server's hostname) []:www.open.net
Email Address []:admin@open.net

Please enter the following 'extra' attributes
to be sent with your certificate request
A challenge password []:
An optional company name []:
```

#ls –l /

Step 3. 개인키와 CSR을 이용해 인증서 생성

#openssl x509 –in www.open.net.csr –out www.open.net.crt –req –signkey www.open.net.key –days 365

```
[root@open certs]# openssl x509 -in open.net.csr -out open.net.crt -req -signkey open.net.key -days 365
Signature ok
subject=/C=KR/ST=Seoul/L=Seoul/O=OpenSouce Corp./OU=Security&System/CN=www.open.net/emailAddress=admin@open.net
Getting Private key
```

❊ 설치된 파일 확인

#ls –l www.open.net.*

```
[root@open certs]# ls -l open.net.*
-rw-r--r--. 1 root root 1334 11월  6 15:44 open.net.crt
-rw-r--r--. 1 root root 1070 11월  6 15:37 open.net.csr
-rw-r--r--. 1 root root 1675 11월  6 15:34 open.net.key
```

Step 4. httpd-ssl.conf 파일에 인증서 경로 설정

vi /etc/httpd/conf.d/httpd-ssl.conf

SSLCertificateFile **/etc/pki/tls/certs/www.open.net.crt**

SSLCertificateKeyFile **/etc/pki/tls/certs/www.open.net.key**

```
 95 #   Server Certificate:
 96 # Point SSLCertificateFile at a PEM encoded certificate.  If
 97 # the certificate is encrypted, then you will be prompted for a
 98 # pass phrase.  Note that a kill -HUP will prompt again.  A new
 99 # certificate can be generated using the genkey(1) command.
100 SSLCertificateFile /etc/pki/tls/certs/open.net.crt
101
102 #   Server Private Key:
103 #   If the key is not combined with the certificate, use this
104 #   directive to point at the key file.  Keep in mind that if
105 #   you've both a RSA and a DSA private key you can configure
106 #   both in parallel (to also allow the use of DSA ciphers, etc.)
107 SSLCertificateKeyFile /etc/pki/tls/certs/open.net.key
```

Step 5. 방화벽 설정

#iptables –I INPUT –m state --state NEW, ESTABLISHED –p tcp --dport 443 -j ACCEPT

#service iptables save

#iptables –L

```
ACCEPT     tcp  --  anywhere   anywhere   state NEW,ESTABLISHED tcp dpt:http
ACCEPT     tcp  --  anywhere   anywhere   state NEW,ESTABLISHED tcp dpt:https
```

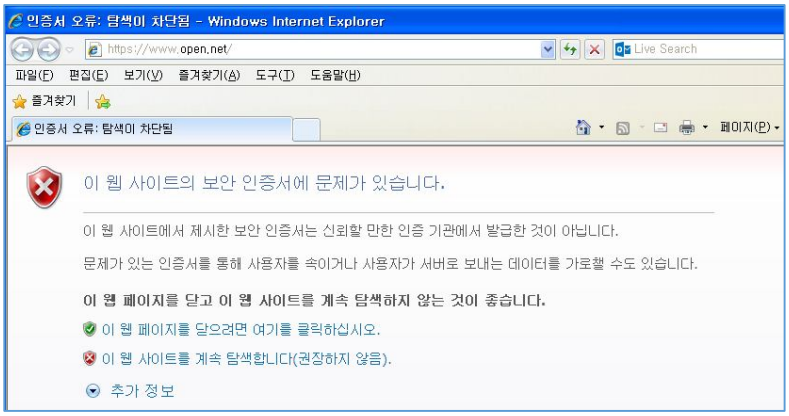
Step 6. 서비스 활성화

#service iptables restart

#service httpd restart

3) HTTPS Client

Step 1. https://www.open.net 접속

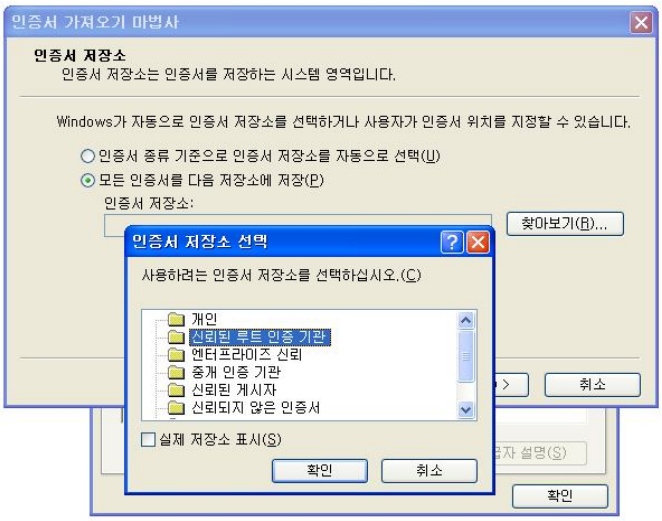


Step 2. 인증서 설치

❶ 인증서 오류 클릭

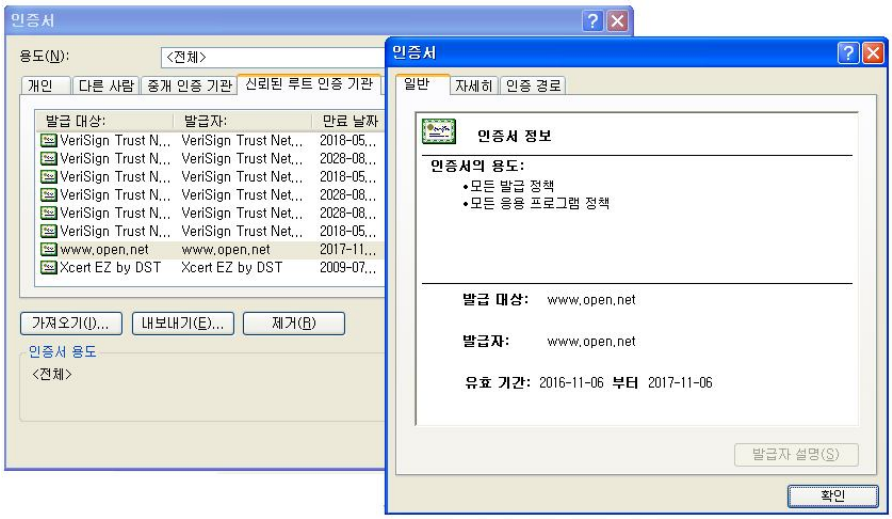
❷ 인증서 설치 클릭

❸ '신뢰된 루트 인증기관'에 인증서 저장



❹ 설치된 인증서 확인

웹브라우저 > 도구 > 인터넷 옵션 > 내용 > 인증서



Step 3. https://www.open.net 재접속